

1 CRITICAL SECTION

Critical perimeter at $d/2$ from the column face, concentrated load, or reaction area.

$$A_c = b_o d \quad (1)$$

A_c area of concrete critical section [in²]
 b_o perimeter of critical section [in]
 d average effective slab depth [in]

2 FACTORED SHEAR STRESS

$$v_u = \frac{V_u}{A_c} + \frac{\gamma_{vx} M_{ux} c_y}{J_{cx}} + \frac{\gamma_{vy} M_{uy} c_x}{J_{cy}} \quad (2)$$

v_u factored two-way shear stress at critical section [psi]
 V_u factored shear transferred across critical section [lb]
 γ_{vx} fraction of M_{ux} transferred by eccentricity of shear
 M_{ux} factored unbalanced moment about x axis [lb-in]
 c_y distance from critical-section centroid to checked perimeter point in y direction [in]
 J_{cx} critical-section property for stress from M_{ux} [in⁴]
 γ_{vy} fraction of M_{uy} transferred by eccentricity of shear
 M_{uy} factored unbalanced moment about y axis [lb-in]
 c_x distance from critical-section centroid to checked perimeter point in x direction [in]
 J_{cy} critical-section property for stress from M_{uy} [in⁴]

2.1 J_c FROM CRITICAL SECTION GEOMETRY

$$J_c = \sum_i (I_i + A_i r_i^2) \quad (3)$$

I_i segment property about its own centroidal axis [in⁴]
 A_i segment area of critical section [in²]
 r_i distance from segment centroid to centroidal axis of critical section [in]

2.2 MOMENT TRANSFER FRACTION

$$\gamma_v = 1 - \gamma_f, \quad \gamma_f = \frac{1}{1 + \frac{2}{3} \sqrt{b_1/b_2}} \quad (4)$$

γ_v fraction of unbalanced moment transferred by eccentricity of shear
 γ_f fraction of unbalanced moment transferred by flexure
 b_1 critical-section width perpendicular to axis of moment [in]
 b_2 critical-section width parallel to axis of moment [in]

3 CONCRETE SHEAR STRENGTH

$$v_c = \min \left\{ \begin{array}{l} 4\lambda_s \lambda \sqrt{f'_c} \\ 2 \left(1 + \frac{2}{\beta} \right) \lambda_s \lambda \sqrt{f'_c} \\ \left(2 + \frac{\alpha_s d}{b_o} \right) \lambda_s \lambda \sqrt{f'_c} \end{array} \right\} \quad (5)$$

$$\lambda_s = \min \left(1.0, \sqrt{\frac{2}{1 + d/10}} \right) \quad (6)$$

v_c nominal two-way shear stress from concrete [psi]
 λ_s size effect factor, d in inches
 f'_c specified concrete compressive strength [psi]
 α_s 40 interior, 30 edge, 20 corner

4 HEADED SHEAR STUDS

4.1 INNER CRITICAL SECTION

$$v_{c,\text{stud}} = \min \left\{ \begin{array}{l} 3\lambda_s \lambda \sqrt{f'_c} \\ 2 \left(1 + \frac{2}{\beta} \right) \lambda_s \lambda \sqrt{f'_c} \\ \left(2 + \frac{\alpha_s d}{b_o} \right) \lambda_s \lambda \sqrt{f'_c} \end{array} \right\} \quad (7)$$

$$v_s = \frac{A_v f_{yt}}{b_o s}, \quad \frac{A_v}{s} \geq \frac{2\sqrt{f'_c} b_o}{f_{yt}} \quad (8)$$

$v_{c,\text{stud}}$ nominal concrete stress at inner stud-reinforced critical section [psi]
 A_v sum of stud areas on one peripheral line [in²]
 f_{yt} yield strength of shear studs used for design [psi]
 s spacing of peripheral stud lines perpendicular to column face [in]

INNER LIMITS

Strength $\phi(v_{c,\text{stud}} + v_s) \geq v_u$
 Compression cap $v_u \leq 8\phi\sqrt{f'_c}$
 f_{yt} used in v_s limited by ACI transverse reinforcement limit

4.2 OUTER CRITICAL SECTION

$$v_{c,\text{out}} = 2\lambda_s \lambda \sqrt{f'_c}, \quad \phi v_{c,\text{out}} \geq v_{u,\text{out}} \quad (9)$$

$v_{c,\text{out}}$ nominal concrete stress beyond outermost stud line [psi]
 $v_{u,\text{out}}$ factored stress at outer critical section using outer A_c , J_c , and c [psi]

4.3 STUD GEOMETRY

$$h_{\text{stud}} \geq h - c_{\text{top}} - c_{\text{rail}} - \frac{d_b}{2} \quad (10)$$

h_{stud} overall height of headed shear stud assembly [in]
 h slab thickness [in]
 c_{top} concrete cover on top flexural reinforcement [in]
 c_{rail} concrete cover on base rail [in]
 d_b flexural tension reinforcement bar diameter [in]
 ℓ_{shaft} smooth stud shaft length [in]

PLACEMENT LIMITS

Stud orientation perpendicular to slab plane
 Column face to first peripheral stud line $\leq d/2$
 Stud-line spacing, $v_u \leq 6\phi\sqrt{f'_c}$ $s \leq 3d/4$
 Stud-line spacing, $v_u > 6\phi\sqrt{f'_c}$ $s \leq d/2$
 Adjacent studs on first peripheral line $\leq 2d$
 Size-effect waiver $\lambda_s = 1.0$ only if $\ell_{\text{shaft}} \leq 10$ in and A_v/s min satisfied

5 STRENGTH CHECK

$$\phi v_n \geq v_u, \quad v_n = v_c + v_s, \quad \phi = 0.75 \quad (11)$$

v_n nominal two-way shear stress [psi]
 v_s nominal two-way shear stress from shear reinforcement [psi]

CHECK ORDER

- 1 Set inner critical perimeter and remove ineffective portions at openings or slab edges
- 2 Compute A_c and J_c from the actual critical section geometry
- 3 Compute v_u using direct shear plus $\gamma_v M_{uc}/J_c$
- 4 If studs are used, check inner v_u cap and design A_v/s
- 5 Set outer critical perimeter at $d/2$ beyond the last stud line
- 6 Recompute outer A_c , J_c , c , and $v_{u,\text{out}}$ with no v_s
- 7 Check stud geometry, first-line distance, line spacing, and stud spacing
- 8 Pass when all inner and outer perimeter points satisfy $\phi v_n \geq v_u$